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Towards robust deep reinforcement learning-based quantitative trading with neuro-symbolic trend analysis

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ABSTRACT

Deep reinforcement learning (DRL) has revolutionized quantitative trading (Q-trading) by achieving decent performance without significant human expert knowledge. Despite its achievements, we observe that the current state-of-the-art DRL models are still ineffective in identifying the market trends, causing them to miss good trading opportunity or suffer from large drawdowns when encountering market crashes. To address this limitation, a natural approach is to incorporate human expert knowledge in identifying market trends. Whereas, such knowledge is abstract and hard to be quantified. In order to effectively leverage abstract human expert knowledge, in this paper, we propose a universal logic-guided deep reinforcement learning framework for Q-trading, called Logic-Q. In particular, Logic-Q adopts the program synthesis by sketching paradigm and introduces a neuro-symbolic trend analysis mechanism (NeSy-TA), which can dynamically determine the market trend and the corresponding tuning parameters for backbone models across different modalities. Extensive evaluations on two popular quantitative trading tasks demonstrate that Logic-Q outperforms the previous state-of-the-art baselines by a large margin, which includes the recently proposed powerful multimodal LLM-based trading strategy.

1. Introduction

Deep reinforcement learning (DRL) has revolutionized many quantitative trading tasks, e.g., stock trading (Ee et al., 2020; Nan et al., 2022), portfolio allocation (Cui et al., 2023; Guan & Liu, 2021), order execution (Fang et al., 2021; Zhang et al., 2023), etc.. Unlike traditional analytical solutions (Almgren & Chriss, 2001; Bertsimas & Lo, 1998), the DRL strategies do not require significant human expert knowledge to design and are able to capture the market's microstructure automatically (Fang et al., 2021).

Despite its great success, previous works present that the DRL strategies are prone to overfitting to spurious noise within the historical context sequence (i.e., training data), which results in their poor performance on testing context sequence (Zhang et al., 2023) when encountering extreme market conditions. To achieve more robust and profitable DRL strategies, many recent efforts have been made. *E.g.*, for the order execution task, Fang et al. (2021) propose a novel policy distillation approach based on the teacher-student learning paradigm, which presents better performance on future trading (test) period, and for the stock trading task, Liu et al. (2022), Yang et al. (2020) introduce a novel ensemble reinforcement learning approach which dynamically selects the best-performing agent policy among three popular DRL policies. This approach has been shown to achieve better risk-adjusted returns during the testing period.

However, even state-of-the-art DRL strategies struggle to accurately identify market trends. This leads to missed trading opportunities and large drawdowns when encountering market crashes. Fig. 1 shows the cumulative return curves of two state-of-the-art DRL stock trading strategies (i.e., Sharpe-Ens Liu et al., 2022; Yang et al., 2020 and AlphaMix Sun et al., 2023) during two market crashes of the US stock market. p_t denotes the Dow Jones Industrial Average (DJIA) market index. Specifically, for both cases, we observe that as p_t descends, both DRL strategies fail to foresee the market crash, and the cumulative return drops significantly. A natural idea to improve the performance of the DRL trading strategies is to embed human expert knowledge (Brock et al., 1992; Zhou et al., 2019) regarding market analysis, as previous works show that the technical indicators of the market are effective in guiding trading decisions (Lo et al., 2000; Stoll & Whaley, 1990).

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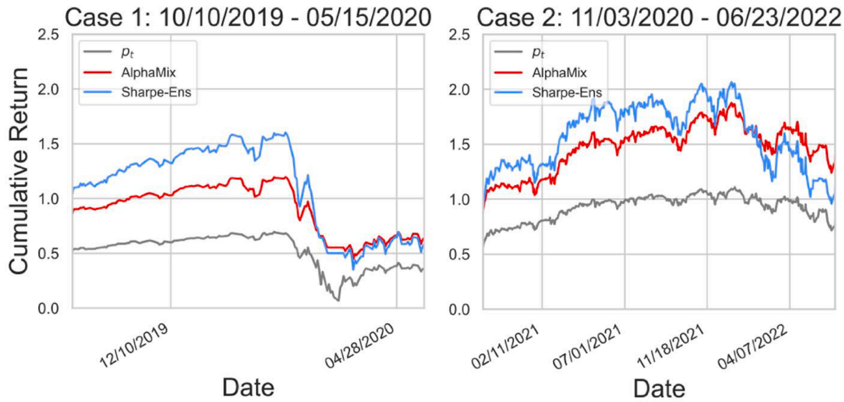


Methods	Case 1			Case 2		
	p_t	AlphaMix	Sharpe-Ens	p_t	AlphaMix	Sharpe-Ens
MD	-37.1%	-34.3%	-68.3%	-18.1%	-22.1%	-48.4%

Fig. 1. Illustration of the cumulative return curves of two state-of-the-art DRL stock trading strategies during two market crashes of the US stock market.

Whereas, the human expert knowledge of market trends is abstract and difficult to quantify because the concrete numeric values associated with market indicators are hard to be manually specified. *E.g.*, the following first-order-logic rule represents a straightforward human understanding of a slow decline market trend:

$$\texttt{slow-decline} \leftarrow (\mathrm{VOL}(t) < \alpha) \wedge (\mathrm{DSR}(t) > \beta) \tag{1}$$

The downside risk indicator DSR(t) of the current time step t measures the potential price decrease, and the volatility indicator VOL(t) measures the degree of variation in the trading price of a market at time step t. A slow decline in the market trend can be identified when the downside risk exceeds a threshold α while the volatility remains below a threshold β, indicating a steady downward price movement without significant fluctuations. However, it is hard to specify the detailed value of α, β as the market changes over time.

To address this challenge and make the best use of the abstract human expert knowledge, in this paper, we adopt the program synthesis by sketching paradigm (Solar-Lezama, 2008) and propose a universal logic-guided deep reinforcement learning framework for Q-trading, called **Logic-Q**.

Specifically, Logic-Q adopts a logic-guided model design that proposes using a neuro-symbolic trend analysis mechanism (NeSy-TA) to embed human expert knowledge of market trends, which consists of two sub-modules, namely the symbolic trend analyzer (Symbolic-TA) and the neural trend analyzer (Neural-TA). Concretely, the Symbolic-TA is essentially a program sketch that embeds abstract human expertise regarding market trends. It is used to generate a coarse yet globally consistent trend prediction based on the current market information. Once it determines the market trend, it returns a market tag and a corresponding tuning parameter, which is used to adjust the action probability distribution of the downstream backbone DRL policies accordingly. Whereas the market categorizations derived by Symbolic-TA are coarse-grained. In order to achieve more fine-grained market trend categorization and therefore a more precise tuning parameter, we leverage the Neural-TA, which takes the high-level yet coarse market tag as input and infers a tuning parameter for refinement based on implicit fine-grained market trend analysis. We then synergistically combine the tuning parameters inferred by both the Symbolic-TA and Neural-TA into a combo tuning parameter and use it to update the action distribution to achieve more robust trading decisions.

Logic-Q can be applied to both the single-model reinforcement learning setting (i.e., a single neural agent learns and makes decisions independently) and the ensemble reinforcement learning setting (i.e., multiple diversified models are trained jointly and their decisions are aggregated for improved performance). To further demonstrate the scalability of Logic-Q, for the ensemble reinforcement learning setting, we introduce a multimodal implementation of Logic-Q, in which the backbone policy set contains sub-policies that are of different modalities. Despite being straightforward, we observe that with the synergistic and flexible combination introduced by NeSy-TA, Logic-Q manages to make the best use of strategies that are of different modalities and network architectures, thus achieving better performance than the current state-of-the-art multimodal baselines, which come with much more complicated, standalone architecture designs.

We conduct extensive experiments on two popular quantitative trading tasks (i.e., order execution and stock trading) for the effectiveness validation. Experimental results show that the neuro-symbolic trend analysis mechanism manages to conduct precise market categorization and thus appropriate refinement/ combination of the backbone strategies accordingly with the tuning parameter, which allows it to significantly outperform previous state-of-the-art baselines by a large margin regarding both cumulative return and maximum drawdown.

This paper is an extension of our conference version (Li et al., 2025a). The new contributions are summarized as follows:

- We extend the original hard-coded trend analysis mechanism to a neuro-symbolic implementation, which achieves more precise market categorization in an implicit manner and therefore better refinement of the backbone policies.
- We extend the original Logic-Q to a multimodal implementation, though being straightforward, we validate that Logic-Q can well combine strategies of different modalities and network architecture based on different categorized market types.
- Extensive results analysis and detailed ablation study validate the effectiveness of our newly proposed NeSy-TA mechanism and the multimodal implementation by outperforming its predecessor and the recently proposed state-of-the-art multimodal baselines.

2. Preliminary

In this section, we first introduce the Markov decision process (MDP), which is the bedrock of deep reinforcement learning, as well as the fundamentals of the two evaluated tasks in our work (i.e., order execution and stock trading).

2.1. Markov decision process

The sequential decision-making problem can be formally modeled as a *Markov Decision Process (MDP)*, represented by a 5-tuple (S, A, R, P, λ). Here, S denotes the state space, A is the action space, $R : S \times A \to \mathbb{R}$ defines the reward function, $P : S \times A \to S$ specifies the state transition dynamics, and $\lambda \in [0, 1)$ is the discount factor that balances immediate and future rewards. At each time step t, the agent observes a state $s_t \in S$ and selects an action $a_t \in A$ according to a stochastic policy $\pi(a_t|s_t)$, which characterizes the probability of taking action a_t given state s_t. The environment then returns a reward $r_t = R(s_t, a_t)$ and transitions to a new state s_{t+1} according to the transition probability $P(s_{t+1}|s_t, a_t)$. This process continues until a terminal time step T, generating a trajectory of experiences $\{(s_t, a_t, r_t)\}_{t=0}^{T}$. The learning objective of reinforcement learning is to discover an optimal policy π^* that maximizes the expected cumulative discounted return:

$$\pi^* = \arg\max_{\pi} \mathbb{E}_{a_t\sim\pi}\left[\sum_{t=0}^{T} \lambda^t r_t\right], \tag{2}$$

where the expectation is taken over the stochasticity of both the policy and the environment.

2.2. Order execution

Order execution (OE) (Cartea et al., 2015) refers to the process of carrying out a financial transaction based on a buy or sell order placed by an investor in the financial markets. The goal of the OE is to maximize the revenue by placing the right amount of orders at each timestep. In this work, we follow the formulation of Fang et al. (2021), given the state representation $\mathbf{s}_t$ of timestep t, a reinforcement learning agent propose a corresponding $a_t \sim \pi(\mathbf{s}_t)$, a_t is discrete and corresponds to the proportion of the target order Q. And the trading volume to be executed at the next time is $q_{t+1} = a_t \cdot Q$, and the optimization objective of policy π is to maximize the expected cumulative discounted rewards of execution $\arg\max_{\pi}\mathbb{E}_{\pi}\left[\sum_{t=0}^{T-1} \gamma^t R_t^{OE}(s_t, a_t)\right]$ where γ is the discount factor. R_t^{OE} is defined as follows:

$$R_t^{OE}(s_t, a_t) = \overbrace{a_t\left(\frac{p_{t+1}}{\tilde{p}} - 1\right)}^{\text{trading profitability}} - \overbrace{\alpha(a_t)^2}^{\text{market impact penalty}} \tag{3}$$

where $\tilde{p} = \frac{1}{T}\sum_{i=0}^{T-1} p_{i+1}$ is the averaged original market price of the whole time horizon. And the reward R_t^{OE} specifies that the agent policy should maximize the volume weighted price advantage (i.e., trading profitability): $a_t\left(\frac{p_{t+1}}{\tilde{p}} - 1\right)$ and minimize the market impact (measured with a quadratic penalty): $-\alpha(a_t)^2$.

2.3. Stock trading

Stock trading (ST) (Markowitz, 1952; Yang et al., 2020) refers to the task that requires a strategy to determine how to adjust a stock portfolio in order to maximize the cumulative return over time. Formally, let $p_t \in \mathbb{R}^D$ be the share price of D stocks at timestep t, $\mathbf{s}_t$ is the state representation that contains market information and the trader's remaining balance b_t. The RL agent should propose action $\mathbf{a}_t \sim \pi(\mathbf{s}_t)$, $\mathbf{a}_t \in \mathbb{R}^D$, which is a vector of actions over D stocks. The action space is defined as $\{-k, \dots, -1, 0, 1, \dots, k\}$, where $-k$ and k represent the maximum number of shares the agent can sell or buy. Finally, the optimization objective of an ST agent policy is to maximize $\arg\max_{\pi}\mathbb{E}_{\pi}\left[\sum_{t=0}^{T-1} \gamma^t R_t^{ST}(s_t, a_t)\right]$, where R_t^{ST} denotes the balance change, formally:

$$R_t^{ST}(s_t, a_t) = \overbrace{p_t^S a_t^S}^{\text{sell change}} - \overbrace{p_t^B a_t^B}^{\text{buy change}} \tag{4}$$

where $p_t^S a_t^S$ represents the balance change due to selling stocks (i.e., sell change), and $p_t^B a_t^B$ denotes the balance change due to buying stocks (i.e., buy change).

3. Methodology

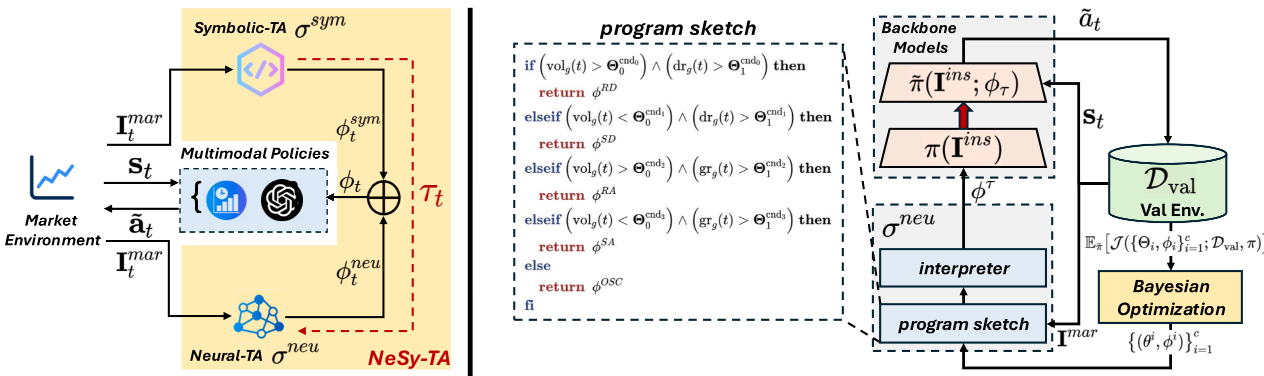
In this section, we introduce the details of the proposed universal logic-guided deep reinforcement learning framework for Q-trading (Logic-Q), which effectively combines human expertise regarding market analysis and neural decision-making for trading. The overview of Logic-Q is shown in Fig. 2.

Concisely, we first introduce a neuro-symbolic trend analysis mechanism (NeSy-TA) to embed abstract human expert knowledge for market analysis. It consists of two sub-modules, namely the symbolic trend analyzer (Symbolic-TA) and the neural trend analyzer (Neural-TA). The Symbolic-TA is essentially a program sketch that embeds abstract human expertise regarding market trends. It is used to generate a coarse yet globally consistent trend prediction based on the current market information. Once it determines the market trend, it returns a market tag and a corresponding tuning parameter, which is used to adjust the action probability distribution of the downstream backbone DRL policies accordingly. Whereas the market categorizations derived by Symbolic-TA are coarse-grained. In order to achieve more fine-grained market trend categorization and therefore a more precise tuning parameter, we leverage the Neural-TA, which takes the high-level yet coarse market tag as input and infers a tuning parameter for refinement based on implicit fine-grained market trend analysis. We then synergistically combine the tuning parameters inferred by Symbolic-TA and Neural-TA into a combo tuning parameter and use it to update the action distribution to achieve more robust trading decisions. The following part of this section is organized as follows: we first illustrate the neuro-symbolic trend analysis mechanism (NeSy-TA) in detail, then we introduce the market trend-based tuning approach, and finally, we detail the proposed optimization method of the holistic Logic-Q framework.

3.1. Neuro-symbolic trend analysis

As aforementioned, the state-of-the-art DRL policies struggle to accurately identify market trends, and the symbolic human expert knowledge regarding market trends is abstract and hard to quantify. In order to effectively embed the human expertise regarding market analysis and therefore boost the robustness of the backbone DRL policies, we adopt the program synthesis by sketching paradigm (Cao et al., 2022; Solar-Lezama, 2008; Verma et al., 2018) and propose a neuro-symbolic trend analysis mechanism (NeSy-TA). As shown in Fig. 2(a), NeSy-TA consists of two major sub-modules, namely symbolic-TA σ^{sym} and neural-TA σ^{neu}. First, for the symbolic-TA σ^{sym}, Fig. 2(b) illustrates its overview and optimization process. Specifically, it is built upon a program sketch (as shown in Fig. 2(b)), which consists of multiple conditional statements, each of them denotes a logical description of a specific type of market trend. The condition of each statement is a boolean expression that logically describes a single market trend. In particular, the program sketch describes five market trends: *steady descend*, *steady ascend*, *rapid descend*, *rapid ascend*, and *oscillation*. And it takes market information $\mathbf{I}^{mar}$ as input, which involves three market indicators: $\mathbf{I}^{mar}(t) = \{\mathrm{vol}_g(t), \mathrm{dr}_g(t), \mathrm{gr}_g(t)\}$, where t denotes the current time step. The details of the three technical indicators are as follows:

- **volatility** ($\mathrm{vol}_g(t) \in [0, +\infty]$) measures the degree of fluctuation of the market price at the timestep t with a history context window of size g, which is calculated as: $\mathrm{vol}_g(t) = \frac{1}{G}\sum_{i=1}^{G}\left(x_i - \bar{x}\right)^2$, where x_i is the market daily close price and $\bar{x}$ is the mean market price of a time window of size G.
- **downside risk** ($\mathrm{dr}_g(t) \in [0, +\infty]$) is defined as $\mathrm{dr}_g(t) = \sqrt{\frac{1}{n} \times \sum_{x_t<\bar{x}}^{n}\left(\bar{x} - x_t\right)^2}$, where n is the total number of observation below the market mean price $\bar{x}$ and x_t is the corresponding



(a) Overview of the Logic-Q framework

(b) The symbolic-TA and its optimization process

Fig. 2. **(a)** Overview of the Logic-Q framework. The NeSy-TA module contains two major sub-modules: the symbolic-TA and the neural-TA; **(b)** The illustration of the symbolic-TA as well as its optimization process.

market close price. It measures the variance of the negative return of the whole trading period. We use it as an indicator to reflect the potential loss of the market.

- **growth rate** ($\mathrm{gr}_g(t) \in [0, +\infty]$) measures the percentage of increase over a specified time span. We use it as an indicator to reflect the market's ascending tendency. Formally,

$$\mathrm{gr}_g(t) = \begin{cases} \left(\frac{X_{\mathrm{t}} - X_{\mathrm{start}}}{X_{\mathrm{start}}}\right) \times 100\% & \text{if } X_{\mathrm{t}} > X_{\mathrm{start}} \\ 0\% & \text{if } X_{\mathrm{t}} \le X_{\mathrm{start}} \end{cases}$$

where X_{t} is the market close price at timestep t and X_{start} is the beginning market price in the period of g.

If a condition is satisfied (i.e., a specific type of market trend τ is identified), the corresponding consequent will be executed, which returns a tuning parameter $\boldsymbol{\phi}_t^{sym}$ of the current time step t.

Take the *steady ascend* market trend for example, if the volatility of the current market is relatively low and the growth rate is high, we can infer that the market is ascending steadily. Whereas, if the volatility and growth rate of the current market are both high, it often denotes a *rapid ascend* market. It is obvious that the definition of lowness and highness is ambiguous. To allow boolean expression with ambiguity, we introduce using the *hole* construct (Cao et al., 2022; Solar-Lezama, 2008), which represents unspecified scalar/tensor parameters as placeholders to be completed. Therefore, for example, the above-mentioned understanding of the *steady ascend* market trend can be quantified with the boolean expression: $\left(\mathrm{vol}_g(t) < \Theta_0^{\mathrm{cnd}_3}\right) \wedge \left(\mathrm{gr}_g(t) > \Theta_1^{\mathrm{cnd}_3}\right)$, where Θ and the corresponding consequent tuning parameter $\boldsymbol{\phi}_t^{sym}$ are holes to be completed (i.e., parameterized). Once completed, they would be returned based upon the program execution conducted by a program interpreter (as shown in Fig. 2). However, it is obvious that the symbolic-TA alone only depicts five general types of market, which are coarse-grained and would therefore lead to suboptimal refinement of the backbone policies. To achieve more fine-grained market categorization, we introduce an additional neural-TA σ^{neu}. Concretely, Neural-TA takes the initial market categorization of Symbolic-TA as guidance, and conducts more fine-grained market categorization in an implicit manner, formally:

$$\begin{aligned} &\boldsymbol{\phi}^{neu} = \sigma^{neu}(\mathbf{I}_t^{mar}, \tau_t; \theta), \\ &\quad \text{where } \tau_t = \sigma^{sym}(\mathbf{I}_t^{mar}; \omega) \end{aligned} \tag{5}$$

where $\mathbf{I}_t^{mar}$ denotes the market information of the current time step t, θ, ω denotes the model parameters of σ^{neu} and σ^{sym} respectively. In this work, we use the multilayer perceptron (MLP) (Rumelhart et al., 1986) to implement σ^{neu}. Intuitively, by leveraging the high-level yet coarse market categorization $\tau_t \in [0, m-1]$ as initial guidance (m denotes the number of hard-coded market types within σ^{sym}), neural-TA outputs the refinement tuning parameter $\phi_t^n \in \mathbb{R}^n$ according to its implicit, fine-grained market categorization (n denotes the number of backbone subpolicies). Then, we adjust the tuning parameters generated by σ^{sym}: $\boldsymbol{\phi}_t^{sym} \in \mathbb{R}^n$ with $\boldsymbol{\phi}_t^{neu} \in \mathbb{R}^n$ and return the final tuning parameter $\boldsymbol{\phi}_t$ which is later used for the backbone subpolicies tuning:

$$\boldsymbol{\phi}_t = \lambda \cdot \boldsymbol{\phi}_t^{sym} + (1-\lambda) \cdot \boldsymbol{\phi}_t^{neu} \tag{6}$$

where λ is a coefficient that controls the weights of the two tuning parameters. We set λ to 0.5 by default.

3.2. Market trend-based policy tuning

Given the final tuning parameter $\boldsymbol{\phi}_t$ and a trained DRL policy $\pi_\epsilon(s_t)$, where ϵ is the model parameter of the DRL agent. We now conduct the post-hoc *market trend-based policy tuning*, which can be formulated as follows:

$$\tilde{\pi}(s_t, \boldsymbol{\phi}_t) = f_{\boldsymbol{\phi}_t} \circ \pi_\epsilon(s_t) \tag{7}$$

where $f_{\boldsymbol{\phi}_t}$ denotes a tuning function that leverages the tuning parameter $\boldsymbol{\phi}_t$ to adjust a trained DRL policy π_ϵ, s_t is the state of time step t, $\tilde{\pi}$ represents the post-tuned policy.

Specifically, in this work, to demonstrate the generality of Logic-Q, we implement two types of tuning functions for two corresponding reinforcement learning paradigms, namely single-model reinforcement learning (Fang et al., 2021; Schulman et al., 2017) and ensemble reinforcement learning (Liu et al., 2022; Yang et al., 2020). For the single-model RL scenario, where a single agent is used for decision-making: let π_ϵ be a trained DRL policy parameterized by ϵ, we freeze the DRL model's weight and implement $f_{\boldsymbol{\phi}_t} \circ \pi_\epsilon(s_t)$ as logit scaling with Softmax temperature (Agarwala et al., 2022). Formally:

$$\tilde{\pi}(s_t; \boldsymbol{\phi}_t) = f_{\boldsymbol{\phi}_t} \circ \pi_\epsilon(s_t) = \frac{\exp\left(\frac{\mathrm{logit}(\pi_\theta(\mathbf{a}_i|s_t))}{\boldsymbol{\phi}_t}\right)}{\sum_{j=1}^k \exp\left(\frac{\mathrm{logit}(\pi_\theta(\mathbf{a}_j|s_t))}{\boldsymbol{\phi}_t}\right)} \tag{8}$$

Intuitively, $\boldsymbol{\phi}_t$ is used as the softmax temperature which dynamically adjusts the action probability distribution according to the determined market trend τ of the current time step t. Specifically, for example, if the pretrained DRL policy performs poorly under the rapid descent market trend, the confidence of the DRL policy should be lowered, thus the corresponding $\boldsymbol{\phi}_t$ would be high: $\boldsymbol{\phi}_t \in (1, +\infty]$, which smooths the action probability distribution.

For the ensemble reinforcement learning scenario, where multiple models are involved for decision-making: let $\Pi = \{\pi_i \mid 0 < i \le k\}$ be a set of subpolicies of size k, the subpolicies are independently trained. Given the observed state $\mathbf{s}_t$ of timestep t, we implement $f_{\boldsymbol{\phi}_t} \circ \pi_\epsilon(\mathbf{s}_t)$ as the bagging-based ensemble and take the weighted average of the predicted action probability distribution of each policy $\pi_i(\cdot \mid s_t; \epsilon_i), 1 \le i \le |\Pi|$, which is as follows:

$$\tilde{\pi}(s_t;\epsilon,\boldsymbol{\phi}_{t,i}) = \sum_{i=1}^{k} \boldsymbol{\phi}_{t,i} \cdot \pi_i(s_t;\epsilon_i),\ 0 < i \le k \tag{9}$$

where $\boldsymbol{\phi}_t$, $||\boldsymbol{\phi}_t||_1 = 1$ is used as the weight tensor for combining the pretrained subpolicies. Intuitively, the NeSy-TA decides the most suitable combination of the subpolicies under different market trends, which allows us to make the best use of different subpolicies. Besides, researchers have recently shown that utilizing multimodal information (e.g., time series, natural language, relational graph, etc.) is beneficial for improving trading performance (Sawhney et al., 2021; Sun et al., 2023). We therefore extend the ensemble reinforcement learning setting of Logic-Q to the multimodal setup. Concretely, in this work, we integrate backbone subpolicies that are of two prevalent financial data modalities, namely time series and natural language. Concretely, to embed natural language information, we adopt the large language models (LLMs) to embed daily stock news and infer the corresponding trading actions to take; and for time series, we use the state-of-the-art RL-based trading strategies, which solely use financial time series as input, formally:

$$\tilde{\mathbf{a}}_t = \boldsymbol{\phi}_{t,0} \cdot \texttt{TS-P}(\mathbf{s}_t^{TS}) + \boldsymbol{\phi}_{t,1} \cdot \texttt{LLM}(\mathbf{s}_t^{NL}) \tag{10}$$

where TS-P denotes the backbone time series-based subpolicy that takes financial time series as input and generate the corresponding trading signals, LLM is the backbone LLM subpolicy that is used to embed natural language information and generate the corresponding trading signals. In this way, Logic-Q can dynamically adjust the weights of the trading signals generated by subpolicies that are of different modalities according to the current market trends. Unlike previous multimodal solutions that require a huge standalone backbone model for heterogeneous data embedding, we could achieve more flexible data fusion by processing multimodal information asynchronously and leveraging the power of different modality-specific backbone models.

3.3. Logic-Q optimization

In this section, we introduce the optimization process of Logic-Q. We illustrate the optimization process under the ensemble reinforcement learning setting of the downstream evaluated stock trading task, the single-model reinforcement learning setting is the same except that it adopts a conventional training-validation-test split instead of a rolling approach. Concretely, to achieve online learning such that our model can better adapt to the changes of the versatile stock markets, we adopt a rolling training-evaluation approach, the details are shown in Algorithm 1. First, the initial training period $\mathcal{E}^{\text{train}}$ is set to be of length η: $\mathcal{E}^{\text{train}} \leftarrow [0, t=\eta)$, and the validation and test periods $\mathcal{E}^{\text{val}}$, $\mathcal{E}^{\text{test}}$ are defined as the two consecutive time intervals of length ρ that follow immediately after:$\mathcal{E}^{\text{val}} \leftarrow [t, t+\rho)$, $\mathcal{E}^{\text{test}} \leftarrow [t+\rho, t+2\rho)$. We first update the backbone time series-based policy TS-P (parameterized by ξ) on the training period by maximizing the optimization objective $\mathcal{J}^{\text{TS-P}}$: $\xi^* = \arg\max_{\xi} \mathbb{E}_{\text{TS-P}}\left[\mathcal{J}^{\text{TS-P}}(\xi;\mathcal{E}^{\text{train}})\right]$, we use Sharpe ratio (Sharpe, 1998) as $\mathcal{J}^{\text{TS-P}}$. Secondly, we optimize the symbolic trend analyzer σ^{sym} parameterized by θ. Concretely, we fix the weights of the backbone policies Π and update σ^{sym} by maximizing the objective $\mathcal{J}^s$ based on the validation period $\mathcal{E}^{\text{val}}$: $\{\theta^*, \phi_s^*\} = \arg\max_{\{\theta,\phi_s\}} \mathbb{E}_{\Pi}\left[\mathcal{J}^s(\{\theta,\phi_s\};\mathcal{D}_{\text{val}},\Pi)\right]$. We adopt Bayesian optimization (Snoek et al., 2012) following (Li et al., 2025b) due to the discrete, non-differentiable nature of the symbolic program sketch. Third, we collect the coarse market categorization tags inferred by σ^{sym}, which are used as high-level guidance for the neural trend analyzer σ^{neu}. We store the inferred results on the training period into a buffer $\mathcal{B}$: $\mathcal{B} \leftarrow \mathcal{B} \cup \{\tau_t\}$. Then, given $\mathcal{B}$, we fix the weights of Π and σ^{sym}, and optimize σ^{neu} by maximizing $\mathcal{J}^n$: $\{\omega^*, \phi^{n,*}\} = \arg\max_{\omega,\phi^n} \mathbb{E}_{\Pi}\left[\mathcal{J}^n(\{\omega,\phi_n\};\mathcal{D}_{\text{train}},\mathcal{B},\Pi)\right]$. Finally, we conduct evaluation on the test period by leveraging the final tuning parameter: $\boldsymbol{\phi}_t = \lambda \cdot \boldsymbol{\phi}_t^{sym} + (1-\lambda)\cdot \boldsymbol{\phi}_t^{neu}$. We roll forward the traning-val-test data by a window length of ρ and evaluate the models' performance recursively.

Algorithm 1: Logic-Q optimization process.

Input: Dataset $\mathcal{D}$, rolling window length ρ, training period length η, weight factor ϵ, trend-buffer $\mathcal{B}$, time series-based polciy TS-P, NL-based policy LLM
Initialize: Anchor $t \leftarrow \eta$
while *not finished* **do**
  // training-validation-test env split
  $\mathcal{E}^{\text{train}} \leftarrow [0, t)$;
  $\mathcal{E}^{\text{val}} \leftarrow [t, t+\rho)$;
  $\mathcal{E}^{\text{test}} \leftarrow [t+\rho, t+2\rho)$;
  // TS-P optimization
  $\xi^* = \arg\max_{\xi} \mathbb{E}_{\text{TS-P}}\left[\mathcal{J}^{\text{TS-P}}(\xi;\mathcal{E}^{\text{train}})\right]$;
  // Symbolic trend analyzer optimization
  $\{\theta^*, \boldsymbol{\phi}^{sym,*}\} = \arg\max_{\{\theta,\boldsymbol{\phi}^{sym}\}} \mathbb{E}_{\Pi}\left[\mathcal{J}^s(\{\theta,\phi^{sym}\};\mathcal{D}_{\text{val}},\Pi)\right]$;
  // Market trend collection
  foreach *timestep t in $\mathcal{E}^{train}$* **do**
    $\tau_t \leftarrow \sigma^{sym}(\mathbf{I}_t^{mar};\theta)$; $\mathcal{B} \leftarrow \mathcal{B} \cup \{\tau_t\}$;
  // Neural trend analyzer optimization
  $\{\omega^*, \boldsymbol{\phi}^{neu,*}\} = \arg\max_{\omega,\boldsymbol{\phi}^{neu}} \mathbb{E}_{\Pi}\left[\mathcal{J}^n(\{\omega,\boldsymbol{\phi}^{neu}\};\mathcal{D}_{\text{train}},\mathcal{B},\Pi)\right]$;
  // Conduct evaluation on the test environment
  foreach *timestep t in $\mathcal{E}^{test}$* **do**
    $\boldsymbol{\phi}_t = \lambda \cdot \boldsymbol{\phi}_t^{sym} + (1-\lambda)\cdot \boldsymbol{\phi}_t^{neu}$;
    $\tilde{\mathbf{a}}_t = \boldsymbol{\phi}_{t,0} \cdot \texttt{TS-Policy}(\mathbf{s}_t^{TS}) + \boldsymbol{\phi}_{t,1} \cdot \texttt{LLM}(\mathbf{s}_t^{NL})$;
    $R_{\mathcal{E}^{\text{test}}} \leftarrow \sum_t r(\mathbf{s}_t, \tilde{\mathbf{a}}_t)$;
  // Roll forward
  $t \leftarrow t + \rho$;
  $\mathcal{E}^{\text{train}} \leftarrow \mathcal{E}^{\text{train}} \cup \mathcal{E}^{\text{val}}$;

4. Experiments

To evaluate the effectiveness of the Logic-Q framework, we conduct experiments on two popular quantitative trading tasks. Concretely, we evaluate the Logic-Q's performance under the single-model reinforcement learning setting on the order execution task; and we evaluate the performance under the ensemble reinforcement learning setting on the stock trading task. In the remainder of this section, we first introduce the detailed experimental setup, and then we answer four research questions (RQs) to lead our discussion, which are as follows:

- **RQ1:** Can Logic-Q effectively improve a single-model reinforcement learning policy's performance?
- **RQ2:** Can Logic-Q effectively improve an ensemble reinforcement learning policy's performance?
- **RQ3:** How does each component of Logic-Q contribute to the model's performance?
- **RQ4:** How well does NeSy-TA identify market trends?

4.1. Order execution setup

4.1.1. Datasets & training

We conduct all experiments on the historical transaction data of the stocks of the China A-shares market provided by Fang et al. (2021). The dataset consists of minute-level intraday price-volume market data of CHINA SECURITIES INDEX 800 (CSI 800) constituent stocks and the order amount of each instrument of each trading day. We follow the official implementation (Fang et al., 2021) and use the data from 2020-09-15 to 2020-11-30 as the training set, 2020-12-01 to 2021-12-31 as the validation set, 2021-01-04 to 2021-01-14 as the test set. For a fair comparison, we use the same training setup as the original implementation (Fang et al., 2021). Specifically, the buffer size is set to be 80,000, the training batch size is set to be 1,024, the learning rate is set to be 0.0001 ,and the maximum training epoch is set to be 30. The maximum number of trials of the Bayesian Optimization is set to be 20. The results of the evaluated methods are all averaged over 5 random seeds (same for the stock trading task).

4.1.2. Compared methods

We compare our method with two traditional financial model-based methods and two state-of-the-art DRL methods proposed for order execution.

- **Time-Weighted Average Price** (TWAP) (Bertsimas & Lo, 1998): TWAP is a rule-based method that equally allocates the target order amount into T pieces over the entire trading time horizon.
- **Volume-Weighted Average Price** (VWAP) (Kakade et al., 2004): VWAP is also a rule-based model that distributes the target order amount to the estimated market transaction volume while monitoring the actual market price.
- **Proximal Policy Optimization** (PPO) (Lin & Beling, 2021): PPO is a policy-based DRL strategy based on an agent with a recurrent neural network and is trained with a sparse reward function.
- **Oracle Policy Distillation** (OPD) (Fang et al., 2021): OPD is a state-of-the-art single-model DRL order execution strategy that utilizes a teacher policy grounded in perfect market information to lead the learning of the target policy.
- **Logic-Q**: we implement Logic-Q by using the OPD strategy as the backbone neural policy to demonstrate the effectiveness of Logic-Q for the single-model RL policy.
- **Logic-Q v0**: the previous Logic-Q version, which only uses the Symbolic-TA instead of the full NeSy-TA.

4.1.3. Evaluation measurements

We evaluate the OE strategies' performance with 4 different measurements:

- **Price advantage** (PA): PA measures the relative gained revenue of a trading strategy compared to a baseline price, formally: $\mathrm{PA} = \frac{10^4}{|\mathbb{D}|}\sum_{k=1}^{|\mathbb{D}|}\left(\frac{p_s^k}{p^k}-1\right)$, where p_s^k is the average execution price that a strategy achieved on order k, p^k is a baseline price (we follow previous work and use the averaged market price of a instrument of a trading day as p^k). PA is measured in basis points (BPs), with one basis point equivalent to 0.01%.
- **Additional annualized rate of return** (ARR): ARR measures the additional annualized rate of return brought by an order execution strategy compared to the TWAP strategy.
- **Gain-loss ratio** (GLR): GLR is a metric that compares the average gain from winning trades to the average loss from losing trades over a trading period, formally: $\mathrm{GLR} = \frac{\mathbb{E}[\mathrm{PA}|\mathrm{PA}>0]}{\mathbb{E}[\mathrm{PA}|\mathrm{PA}<0]}$.
- **Positive rate** (POS): POS measures the positive rate of PA across all orders over a trading period, formally: $\mathbb{E}[\mathrm{PA} > 0]$.

4.2. Stock trading setup

4.2.1. Datasets & training

For the stock trading task, we conduct experiments on the United States stock market, Hong Kong stock market, and the cryptocurrency market. We use the public dataset collected by Yahoo Finance[1]. For the US market, we follow (Yang et al., 2020) and use the Dow Jones 30 constituent stocks as our trading stock pool, and for the HK market, we use Hang Seng China 50 Index constituent stocks as the trading stock pool. For the cryptocurrency market, we select cryptocurrencies that are of top-10 market volume, including BTC/USD, ADA/USD, FIL/USD, ETH/USD, LTC/USD, BNB/USD, EOS/USD, ETC/USD, LINK/USD, and BCH/USD. Because the cryptocurrency market trades 24 hours a day, unlike the Hong Kong and U.S. markets, we set an 8-hour trading interval. For HK and USA market, We use the daily data from 2009-01-01 to 2015-09-26 as the in-domain training data and the data from 2015-09-26 to 2023-06-01 for validation and testing. And for cryptocurrency market, We use 8-hours interval data from 2018-01-01 to 2020-09-25 as the in-domain training data and the data from 2020-09-26 to 2024-05-01 for validation and testing. To avoid survivorship bias, we use the actual index constituent lists as of each rebalancing date and dynamically adjust the universe for additions and deletions, excluding any delisted stocks from all future trading windows. Concretely, we follow (Yang et al., 2020) and use a rolling training-validation-test approach. We recursively update the training-validation-test time step by 3 months for the US and HK markets, and by 1 month for the cryptocurrency market, retraining all sub-policies during this period.

The initial portfolio value is set at $1,000,000, with transaction costs at 0.1% of the value of each trade (both buy and sell). The maximum trading volume for a single stock in one day is capped at 100 shares, and for a cryptocurrency within an 8-hour interval, it is limited to $100,000. Idle cash earns zero interest under a conservative assumption, while only borrowed funds in margin trading incur interest at the specified rates. This design ensures that returns are not artificially inflated by cash holdings (Sammon & Shim, 2026). The market indices are replicated as equal-weighted portfolios of their current constituents and rebalanced quarterly to precisely match official index reconstitution events, serving as the passive buy-and-hold benchmark (Lo et al., 2000).

We evaluate Logic-Q and the baselines under both the cash trading and the margin trading scenario. For the margin trading setting, we borrow funds equal to the total value of our account, creating a 1:1 loan-to-value ratio. The borrowing amount is adjusted every three months to maintain the same leverage ratio and to repay the interest incurred from borrowing. The interest is calculated based on the borrowed amount and the agreed-upon interest rate with Robinhood[2] and Binance[3]. Specifically, for the U.S. and Hong Kong stock markets, we impose an annual margin interest rate of 7.75% for Gold plan subscribers, whereas for the cryptocurrency market, we adopt the Binance VIP3 rate, averaging an annual margin interest of 17.12%. Our state representation $s \in \mathbb{R}^{331}$ contains 9 major indicators, namely, available balance at the current time step, adjusted close price of each stock, shares owned of each stock, Moving Average Convergence Divergence (`macd`, `macds`), Bollinger Bands (`boll_ub`, `boll_lb`), Relative Strength Index (`rsi_30`), Commodity Channel Index (`cci_30`), Directional Index(`dx_30`), and Simple Moving Average (`close_30_sma`, `close_60_sma`). As shown in Table 1, all technical indicators are computed exclusively using historical data up to timestep t. Each indicator's computation window is bounded by $[t-n, t]$ for lookback period n, ensuring that only past and current observations are accessible when making trading decisions at time t, thus preventing any look-ahead bias.

4.2.2. LLM-based subpolicy configuration

We incorporate a Large Language Model (LLM) as an additional subpolicy to process real-time news sentiment for trading decisions. Specifically, we use GPT-4o mini (gpt-4o-mini-2024-07-18) accessed via the official OpenAI API for deterministic action generation. News articles are collected in real-time from Yahoo Finance[4] RSS feeds for every asset in our trading universe. For each trading period t, we retrieve only articles published within the preceding 24 hours (for stocks) or 8 hours (for cryptocurrencies). Relevance is determined by keyword matching

[1] https://github.com/yahoo-finance

[2] https://robinhood.com/us/en/support/articles/margin-overview/

[3] https://www.binance.com/en/fee/marginFee

[4] https://github.com/yahoo-finance

Table 1
Technical indicators and their computation using only historical data.

Indicator	Code	Computation at Time t (Historical Data Only)
Simple Moving Average	close_30_sma close_60_sma	$\mathrm{SMA}_{30}(t) = \frac{1}{30}\sum_{i=t-29}^{t} p_i$ $\mathrm{SMA}_{60}(t) = \frac{1}{60}\sum_{i=t-59}^{t} p_i$
Moving Average Convergence Divergence	macd macds	$\mathrm{MACD}(t) = \mathrm{EMA}_{12}(t) - \mathrm{EMA}_{26}(t)$ $\mathrm{MACD}_{\mathrm{signal}}(t) = \mathrm{EMA}_9(\mathrm{MACD}(t))$
Bollinger Bands	boll_ub boll_lb	$\mathrm{BB}_{\mathrm{upper}}(t) = \mathrm{SMA}_{20}(t) + 2 \cdot \sigma_{20}(t)$ $\mathrm{BB}_{\mathrm{lower}}(t) = \mathrm{SMA}_{20}(t) - 2 \cdot \sigma_{20}(t)$
Relative Strength Index	rsi_30	$\mathrm{RSI}_{30}(t) = 100 - \frac{100}{1+\frac{\mathrm{Avg\ Gain}_{30}(t)}{\mathrm{Avg\ Loss}_{30}(t)}}$
Commodity Channel Index	cci_30	$\mathrm{CCI}_{30}(t) = \frac{\mathrm{TP}(t)-\mathrm{SMA}_{30}(\mathrm{TP}(t))}{0.015\cdot \mathrm{MD}_{30}(t)}$
Directional Index	dx_30	$\mathrm{DX}_{30}(t) = 100 \cdot \frac{\lvert \mathrm{DI}^{+}_{30}(t)-\mathrm{DI}^{-}_{30}(t)\rvert}{\mathrm{DI}^{+}_{30}(t)+\mathrm{DI}^{-}_{30}(t)}$

on ticker symbols, company names, and sector terms; only the title, RSS description, and timestamp of matching articles are retained.

The LLM prompt follows a structured two-part design:

- **System message**: "You are a professional quantitative trader analyzing financial news to make trading decisions."
- **User message**: "Based on the following news about {ticker} published in the last {time_window}: [News 1: {Title} - {Description} ({Timestamp}); News 2: {Title} - {Description} ({Timestamp}); …], analyze market sentiment and recommend ONE action: Strong Buy (SB), Buy (B), Hold (H), Sell (S), or Strong Sell (SS). Output format: [Action]."

Here, {ticker} is the asset symbol (e.g., AAPL, BTC-USD), and {time_window} is "24 hours" for stocks or "8 hours" for cryptocurrencies. The five discrete actions map to trading volumes: SB/SS execute maximum allowed shares ($\pm k$), B/S execute half ($\pm k/2$), and H maintains current position (0).

To strictly ensure no look-ahead bias and perfect temporal alignment, we filter all news strictly by publication timestamp and guarantee that at any time step t (during both training and rolling evaluation), the model receives only news published before the market close of period t. No future information is ever accessible. The average inference latency is 1.8 s per asset using batch processing with concurrent API calls, which is well within our trading frequency and poses no practical constraint.

4.2.3. Compared methods

We compare our methods with the following baseline stock trading methods.

- **Stock Market Index** (p_t): A stock market index is a measure that reflects the performance of the stock market. It is calculated by aggregating the value of a selected group of stocks. We use the Dow Jones Industrial Average (DJIA) for the United States market and we use the Hang Seng China 50 Index (HSI50) for the Hong Kong market (So & Tse, 2004).
- **Buy and Hold** (BAH): which is commonly employed to represent the market average and signifies a trading approach where predetermined financial assets are purchased in full at the outset and held until the conclusion of the trading period.
- **Deep Deterministic Policy Gradient** (DDPG) (Lillicrap et al., 2015): DDPG is an off-policy DRL algorithm that combines Q-learning (Sutton & Barto, 2018) with the policy gradient methods (Sutton et al., 1999).
- **Advantage Actor Critic** (A2C) (Mnih et al., 2016): A2C is an actor-critic-based DRL algorithm that leverages an advantage function to reduce the variance of policy gradient.
- **Proximal Policy Optimization** (PPO) (Schulman et al., 2017): PPO is a DRL algorithm that optimizes the policy function using the trust region policy optimization approach. It employs a clipped surrogate objective to prevent large policy updates.
- **Sharpe-Ens** (Yang et al., 2020): (Yang et al., 2020) propose this novel ensemble strategy (denoted as Sharpe-ensemble in this paper) which dynamically selects the best-performed DRL agent among a set of candidates according to their Sharpe ratio (Sharpe, 1994) on the validation data.
- **AlphaMix** (Sun et al., 2023): (Sun et al., 2023) introduce an efficient ensemble learning method to train multiple trading experts with different market trend capabilities, conduct model augmentation from a diversified expert pool, and leverage a three-stage mixture of experts framework to dynamically routing experts from the pool.
- **FinAgent** (Zhang et al., 2024). FinAgent is a multimodal agent that leverages transformer architectures to process numerical, textual, and visual financial data for trading decisions, achieving performance improvements across multiple benchmark datasets.
- **Logic-Q**: we implement Logic-Q by using a variant of the Sharpe-Ens strategy as the backbone neural policy[5] to demonstrate the effectiveness of Logic-Q for the ensemble RL policy.
- **Logic-Q v0**: the previous Logic-Q version, which only uses the Symbolic-TA instead of the full NeSy-TA, and time series-based backbone policy only instead of both the time series-based subpolicy and LLM subpolicy.

4.2.4. Evaluation measurements

We evaluate the ST strategies' performance with 5 different measurements:

- **Annualized return** (AR): Annualized return is a measure of the average annual rate of return on an investment over a specified period. It is calculated by taking the total percentage return of the investment and dividing it by the number of years the investment was held.
- **Cumulative return** (CR): Cumulative return is the total amount of return on an investment over a specific period. It is calculated by subtracting the initial investment amount from the final investment value.
- **Annualized volatility** (AV): Annualized volatility is a measure of the degree of variation of an investment's returns over a specific period, expressed as an annualized percentage. It is calculated by taking the standard deviation of the investment's returns over the period and multiplying it by the square root of the number of trading days in a year.
- **Maximum drawdown** (MD): Maximum drawdown is a measure of the largest percentage decline in the value of an investment from its peak to its trough over a specific period.

[5] We use the bagging-based ensemble to combine the action probability distributions derived by the backbone subpolicies.

Table 2
The comparative results of different methods on the order execution task.

Method	GLR	POS	PA	ARR
TWAP	0	0	0	0
VWAP	0.93	0.53	3.40	8.94
PPO	0.91 ± 0.01	0.53 ± 0.01	3.27 ± 0.25	8.59 ± 0.63
OPD	0.91 ± 0.01	0.52 ± 0.01	3.41 ± 0.22	8.97 ± 0.56
OPD (Aug)	0.90 ± 0.00	0.51 ± 0.00	3.07 ± 0.21	8.15 ± 0.55
Logic-Q w/o Symbolic-TA	0.92 ± 0.01	0.52 ± 0.01	3.50 ± 0.29	9.22 ± 0.73
Logic-Q v0	0.96 ± 0.00	0.55 ± 0.00	4.33 ± 0.25	11.53 ± 0.63
Logic-Q (Ours)	**1.05** ± 0.01	**0.57** ± 0.00	**5.91** ± 0.24	**14.70** ± 0.69
Improvement over SOTA	**9.4%**	**3.6%**	**36.4%**	**27.5%**

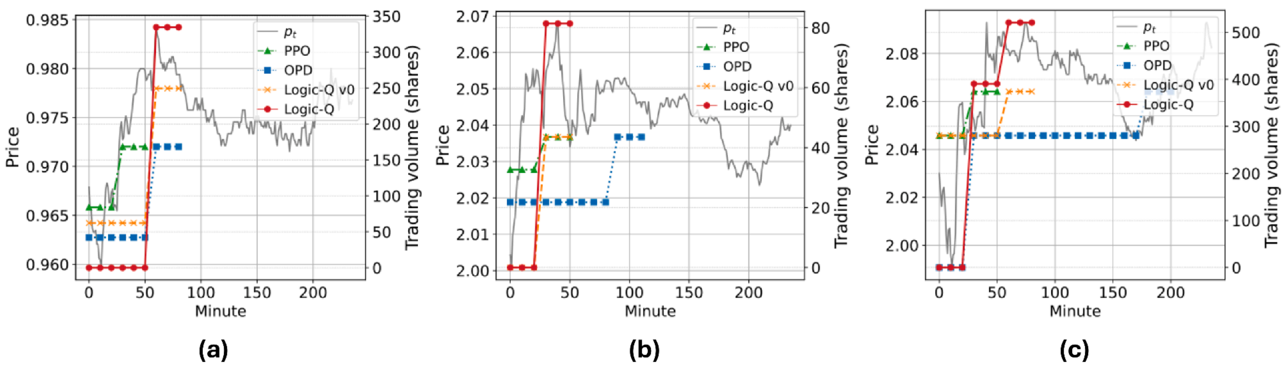


Fig. 3. Case studies of the execution details of different methods.

- **Sharpe ratio** (SR) (Sharpe, 1994): Sharpe ratio is a measure of risk-adjusted return that compares the excess return of an investment over the risk-free rate to its volatility.

4.2.5. Single-model RL improvement (RQ1)

We first evaluate whether Logic-Q can effectively improve the performance of a single-model RL policy on the order execution task.

The results are shown in Table 2. It can be observed that our method substantially increases total returns and reduces maximum drawdowns while maintaining volatility within an acceptable range compared to the previous SoTA OPD method as well as Logic-Q v0, which validates the effectiveness of the proposed NeSy-TA module. To demonstrate the imperativeness of the Symbolic-TA module, we conduct an ablation study. Concretely, we remove the Symbolic-TA of Logic-Q (denoted as Logic-Q w/o Symbolic-TA), which degenerates the model into a simple policy with softmax temperature scaling using Bayesian optimization. For a fair comparison, we use the same training setup for the ablation. Table 2 shows that compared with Logic-Q, the performance of Logic-Q w/o Symbolic-TA significantly drops by all metrics, which demonstrates that the Symbolic-TA is effective in embedding human expertise regarding market trends. Besides, to demonstrate the improvement is brought by the model design instead of merely the market information, we augment the state-of-the-art DRL baseline (OPD) by concatenating the market features with the input state (i.e., $[\mathbf{s}_t, \mathbf{T}^{mar}]$) and follow the same training setup, denoted as OPD (Aug) (as shown in Table 2). The results validate that the market information alone is ineffective in boosting the model's performance.

We further conduct an action analysis of different methods. Fig. 3 shows the order execution (selling) details (i.e., to sell a specific number of shares) of different methods on 3 different assets of 3 different days. The colored lines represent the volume of shares traded (i.e., sold) by different strategies per minute, the grey line is the market price p_t of the target trading asset. We can observe that Logic-Q manages to capture better opportunities than all the compared baseline trading strategies. For example, in Case 1, Logic-Q can well identify an incoming market ascent, it holds the shares and sells them only at the highest price of the day, while the other strategies fail to do so and sell at much lower prices.

4.2.6. Ensemble RL improvement (RQ2)

We then study whether Logic-Q is effective in improving the performance of an ensemble RL policy. We conduct experiments on the stock trading task. The results on US and HK stock markets, as well as cryptocurrency market are shown in Table 3, and the detailed cumulative return curves on the US market are shown in Fig. 6. We can see that compared with the state-of-the-art baselines (i.e., Sharpe-Ens, AlphaMix, and Logic-Q v0), Logic-Q manages to achieve a much higher return while decreasing the maximum drawdown, which results in its highest Sharpe ratio.

We also evaluate the performance of the state-of-the-art baseline models with market information augmentation (i.e., AlphaMix (Aug) and Sharpe-Ens (Aug)). The results show that augmenting the models' input with market information is ineffective in improving these models' performance. Furthermore, Logic-Q also outperforms the recently proposed powerful multimodal strategy (i.e.,FinAgent), demonstrating its superiority as a more flexible and effective multimodal combination framework compared to the baseline, which relies on a complex, standalone architectural design.

To demonstrate that Logic-Q's performance gains stem from its neuro-symbolic design rather than merely increased computational complexity, we conduct a comprehensive efficiency analysis on the US stock market. We compare the total execution time of Logic-Q against Sharpe-Ens and FinAgent on identical hardware using a single NVIDIA RTX 3090 GPU under the same experimental setup. We evaluate the efficiency using wall-clock time, all the reported results are averaged over 5 random seeds. Concretely, Logic-Q completes execution in 7.8 ± 0.37 mins, which remains competitive with baselines: only 10.9% slower than Sharpe-Ens (6.95 ± 0.31 mins), and faster than FinAgent (11.2 ± 0.85 mins). The additional time overhead comes from the two lightweight components of Logic-Q: the Bayesian optimization of Symbolic-TA takes up 0.57 ± 0.06 mins and Neural-TA training takes up 0.33 ± 0.08 mins. Notably, the LLM-based components operate in pure inference mode with zero training cost, adding only 3 to 5 s per

Table 3
The comparative results of different methods on the stock trading task.

US stock market										
Methods	Cash Trading					Margin Trading				
	AR	CR	AV	MD	SR	AR	CR	AV	MD	SR
p_t	9.3%	91.7%	18.9%	-38.0%	0.564	9.3%	91.7%	18.9%	-38.0%	0.564
DDPG	9.3%	82.7%	19.6%	-31.7%	0.585	15.0%	164.8%	36.2%	-49.4%	0.587
A2C	9.7%	94.7%	17.2%	-25.2%	0.599	21.4%	303.3.%	26.7%	-31.9%	0.851
PPO	11.5%	121.7%	16.9%	-22.7%	0.807	10.3%	105.6%	26.8%	-27.9%	0.549
AlphaMix	14.2%	169.7%	20.3%	-33.8%	0.816	23.4%	372.6%	29.9%	-34.3%	0.863
Sharpe-Ens	10.3%	124.2%	33.7%	-68.6%	0.509	22.5%	332.9%	40.2%	-61.5%	0.736
AlphaMix (Aug)	15.4%	181.9%	20.2%	-38.1%	0.831	21.9%	370.5%	23.2%	-37.0%	0.786
Sharpe-Ens (Aug)	11.4%	131.2%	30.4%	-52.7%	0.548	21.3%	318.8%	36.6%	-50.8%	0.662
FinAgent	12.6%	139.8%	17.4%	-24.6%	0.849	22.4%	341.7%	29.5%	-47.2%	0.836
Logic-Q v0	18.1%	239.6%	16.5%	-22.3%	1.097	32.9%	697.9%	27.1%	-26.1%	1.175
Logic-Q (Ours)	**21.2%**	**311.8%**	**16.1%**	**-21.8%**	**1.280**	**38.1%**	**965.3%**	26.9%	**-24.6%**	**1.272**
Improvement over SOTA	**17.1%**	**30.1%**	**2.5%**	**2.3%**	**16.7%**	**15.8%**	**38.3%**	-	**7.3%**	**8.3%**

HK stock market										
Methods	Cash Trading					Margin Trading				
	AR	CR	AV	MD	SR	AR	CR	AV	MD	SR
p_t	-0.1%	-5.4%	21.3%	-55.7%	0.069	-0.1%	-5.4%	21.3%	-55.7%	0.069
DDPG	0.5%	3.7%	20.2%	-52.9%	0.131	0.8%	20.7%	57.9%	-81.6%	0.346
A2C	4.9%	35.9%	18.1%	-42.6%	0.347	4.9%	41.2%	33.8%	-61.3%	0.352
PPO	4.2%	39.4%	24.1%	-51.4%	0.332	1.8%	15.1%	39.0%	-68.6%	0.245
AlphaMix	5.3%	46.4%	19.4%	-40.7%	0.473	16.7%	223.0%	34.1%	-51.9%	0.727
Sharpe-Ens	0.8%	8.0%	38.3%	-75.4%	0.184	11.6%	125.5%	42.7%	-56.8%	0.483
AlphaMix (Aug)	5.3%	49.9%	20.3%	-38.7%	0.465	19.2%	234.6%	34.5%	-47.8%	0.926
Sharpe-Ens (Aug)	1.2%	9.0%	29.7%	-68.9%	0.287	14.6%	131.7%	41.6%	-63.7%	0.586
FinAgent	3.4%	29.7%	28.6%	-45.7%	0.277	10.7%	107.2%	35.1%	-59.2%	0.453
Logic-Q v0	11.8%	109.3%	21.3%	-31.9%	0.624	24.1%	411.7%	29.7%	-30.9%	1.135
Logic-Q (Ours)	**12.5%**	**135.2%**	20.8%	**-27.5%**	**0.698**	**29.3%**	**601.5%**	**27.9%**	**-28.2%**	**1.318**
Improvement over SOTA	**5.9%**	**23.5%**	-	**16.0%**	**11.9%**	**22.0%**	**46.1%**	-	**5.1%**	**16.1%**

Cryptocurrency market										
Methods	Cash Trading					Margin Trading				
AV	MD	SR	AR	CR	AV	MD	SR			
BAH	28.7%	228.4%	65.7%	-80.9%	0.718	28.7%	228.4%	65.7%	-80.9%	0.718
DDPG	28.5%	136.1%	68.7%	-77.9%	0.709	0%	0%	267.7%	-100%	0
A2C	25.3%	117.0%	56.7%	-71.5%	0.682	39.7%	212.3%	103.1%	-87.9%	0.844
PPO	46.7%	268.9%	64.8%	-76.3%	0.915	63.5%	433.8%	76.5%	-88.3%	1.020
AlphaMix	54.4%	339.1%	47.4%	-50.3%	1.162	87.4%	750.9%	71.5%	-63.2%	1.242
Sharpe-Ens	20.9%	91.1%	60.2%	-69.6%	0.619	44.1%	278.5%	68.6%	-79.5%	0.842
AlphaMix (Aug)	56.4%	376.9%	49.8%	-47.9%	1.157	86.8%	736.4%	70.9%	-65.1%	1.230
Sharpe-Ens (Aug)	22.7%	100.6%	60.6%	-71.5%	0.643	45.7%	301.5%	69.1%	-80.3%	0.879
FinAgent	32.1%	176.3%	58.8%	-53.8%	1.096	50.2%	352.7%	74.2%	-79.1%	0.837
Logic-Q v0	63.6%	437.9%	42.7%	-46.1%	1.268	92.9%	836.6%	70.6%	-42.8%	1.291
Logic-Q (Ours)	**70.4%**	**583.2%**	**41.4%**	**-45.6%**	**1.357**	**101.7%**	**984.7%**	**69.3%**	**-41.2%**	**1.474**
Improvement over SOTA	**10.7%**	**33.2%**	**3.1%**	**1.1%**	**7.0%**	**9.5%**	**17.7%**	-	**3.9%**	**14.2%**

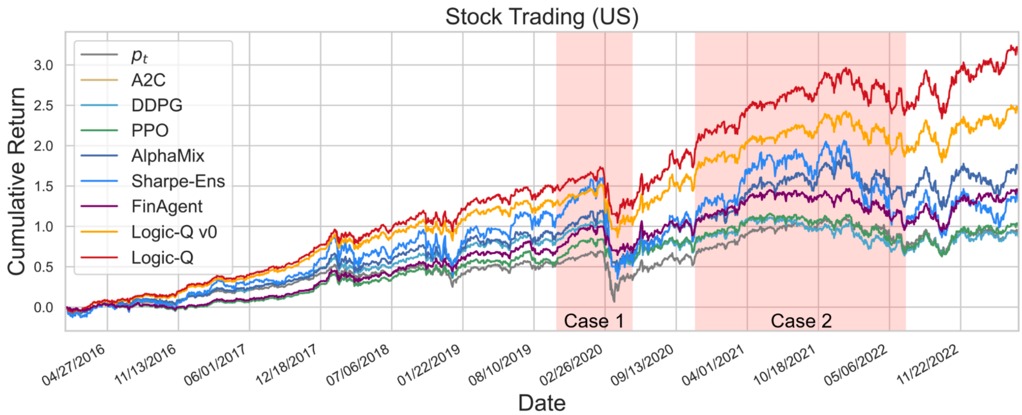
Table 4
Ablation analysis on the US market under the cash trading setting.

Methods	AR	CR	AV	MD	SR
w/o Symbolic-TA, LLM	11.1%	125.0%	16.7%	-27.9%	0.756
w/o LLM	14.8%	172.3%	19.2%	-29.8%	0.771
w/o Symbolic-TA	17.8%	212.4%	22.1%	-32.7%	0.805
w/o Neural-TA	18.6%	221.7%	23.4%	-35.6%	0.795
Logic-Q (Ours)	21.2%	311.8%	16.1%	-21.8%	1.280

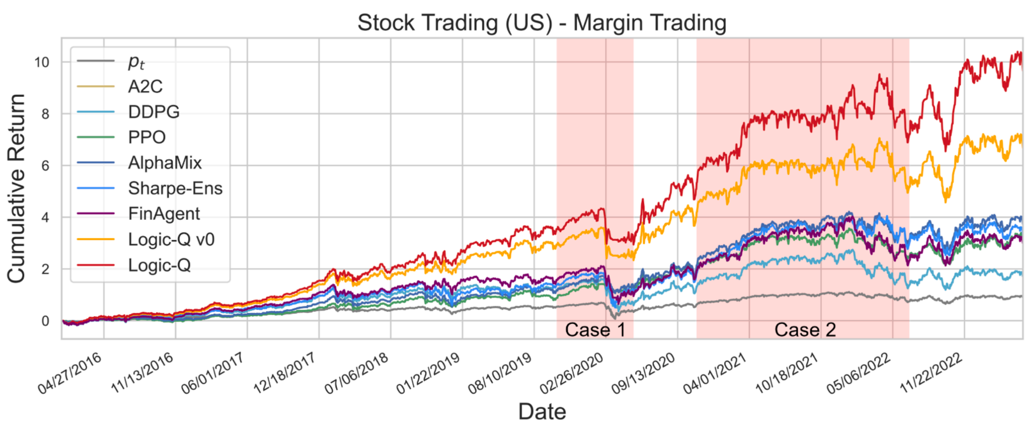
trading day. Our NeSy-TA module introduces minimal parameter overhead: the Neural-TA MLP contains only 0.74K parameters, while the Symbolic-TA adds merely 25 tunable parameters, which consist of 10 threshold parameters and 15 tuning parameters. This negligible parameter increase demonstrates the effectiveness and efficiency of our neural-symbolic design, which achieves substantial performance improvements without significantly increasing model complexity.

4.2.7. Ablation analysis (RQ3)

To understand the contributions of each component of Logic-Q and validate the robustness of our method, we conduct comprehensive ablation studies on the stock trading task in the US market under the cash trading setting. Concretely, we ablate the Symbolic-TA, Neural-TA modules, as well as the LLM backbone subpolicy. , As shown in Table 4, we can observe that the removal of Neural-TA results in a significant cumulative return drop (−79.3%) as well as a nontrivial increase in terms of maximum drawdown(+10.5%). The removal of Symbolic-TA results in a further cumulative return drop (−88.6%), which demonstrate that both the Symbolic-TA and Neural-TA of the NeSy-TA modules contributes the performance gain of the Logic-Q model by effectively embedding human expertise regarding market analysis. We then remove the backbone LLM subpolicy, which results in a significant performance drop (−128.7%@CR, +4.70%@MD). The results demonstrate that the signals derived by the LLM's analysis are effective and contribute to the performance boost. Note that we also present the results of the pure bagging-based ensemble strategy (denoted as w/o Symbolic-TA, w/o LLM). The



(a) Cumulative return curve of different methods for **cash trading** in the US market.



(b) Cumulative return curve of different methods for **margin trading** in the US market.

Fig. 4. Cumulative return curves of different methods on the US market.

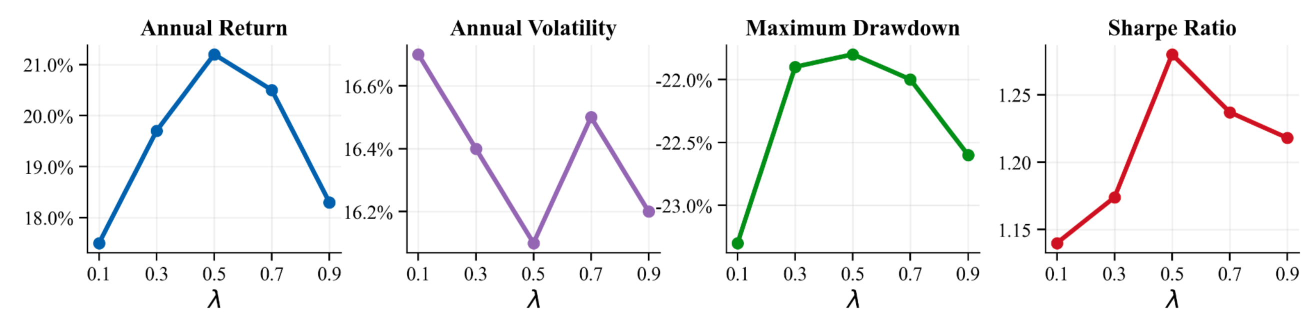


Fig. 5. Sensitivity Analysis of λ.

Table 5
Performance under different transaction costs on US stock market (cash trading).

Transaction Cost	AR	CR	AV	MD	SR
Logic-Q (default)	21.2%	311.8%	16.1%	-21.8%	1.280
Logic-Q (0.13%)	20.6%	295.2%	16.4%	-22.4%	1.238
Logic-Q (0.08%)	21.8%	328.4%	15.8%	-21.3%	1.315

results show that its performance is similar to that of Sharpe-Ens which ensures that the performance boost of Logic-Q is not because of the different ensemble approach.

We further analyze the hyperparameter λ in Eq. (6) that controls the weighted combination between Symbolic-TA and Neural-TA. As illustrated in Fig. 5, $\lambda = 0.5$ yields optimal performance with 21.0% annual return and 1.25 Sharpe ratio in terms of all the measurements, which validates the imperativeness of both Symbolic-TA and Neural-TA.

We also evaluate the performance under different transaction cost settings. Concretely, we evaluate two settings adopted by previous literature: 130 basis points, representing modern low-cost institutional execution (Chen & Velikov, 2023), and 80 basis points, following the effective trading cost estimates for U.S. equities (Hasbrouck, 2009). The results are presented in Table 5. As transaction costs rise, Logic-Q exhibits graceful performance degradation with only modest declines in returns and Sharpe ratio, while volatility and maximum drawdown show marginal increases. This demonstrates that Logic-Q can effectively limit portfolio turnover, preserving the majority of performance advantages even under elevated cost scenarios.

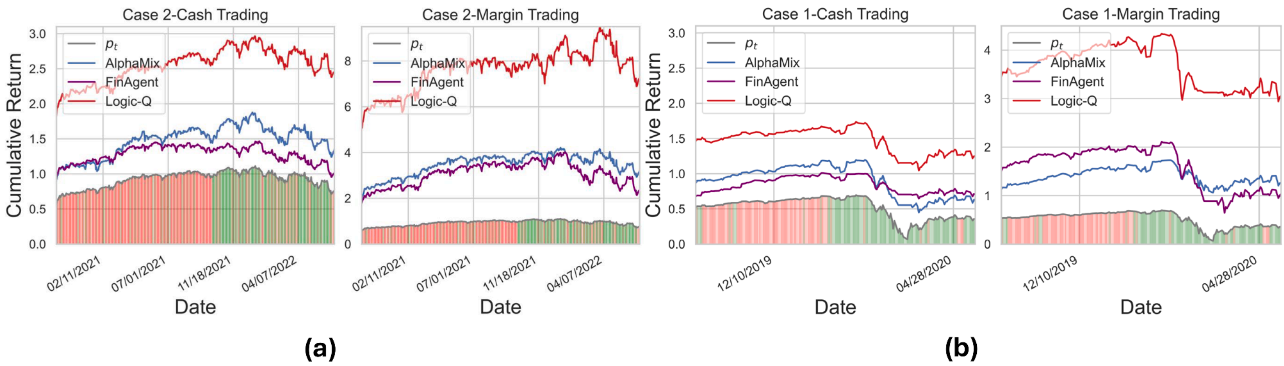


Fig. 6. Cumulative return comparison of different methods under two market descent conditions.

Table 6
Maximum drawdown (%) of different methods during two market crash cases in US Market. CT and MT denote cash trading and margin trading respectively.

Method	Case 1		Case 2	
	CT	MT	CT	MT
p_t	−37.1%	−37.1%	−18.1%	−18.1%
AlphaMix	−34.3%	−29.6%	−22.1%	−29.6%
FinAgent	−24.2%	−47.2%	−20.7%	−37.6%
Logic-Q (Ours)	−21.3%	−21.2%	−14.8%	−24.4%

4.2.8. *Analysis of NeSy-TA (RQ4)*

To better understand the source of improvement, we further conduct an in-depth analysis of the NeSy-TA module. Specifically, we first investigate whether the market trends identified by the Symbolic-TA aligns well with human expertise. We conduct the interpretability analysis based on two cases: (a) a significant market crash case that happened from 03/09/2020 to 03/23/2020, due to the Fed's emergency rate cut that sparked a market-wide fire sale; (b) a slight market drawdown period from 02/11/2021 to 04/07/2022. The results are shown in Fig. 6 as well as the highlighted sections in Fig. 4.

Concretely, for each time step, if the optimized program sketch identifies the current market as an ascending market trend (i.e., rapid ascend or steady ascend), we denote it using a red vertical bar beneath the market price curve p_t, and if the program sketch identifies the current market as a descending trend (i.e., rapid descend or steady descend), we denote it using a green bar. Otherwise, if the program sketch identifies the market as oscillation, we denote it using a white bar. We can observe that for the market crash under both the cash trading and margin trading settings, the area under the ascending period is reddish, and the optimized program sketch manages to timely foresee an incoming market fall (greenish area). The quantitative results are shown in Table 6. We can observe that Logic-Q manages to achieve much lower drawdown compared to the state-of-the-art baselines.

Thus, the overall market trends identified by the Symbolic-TA are interpretable by being strongly aligned with human judgment, which allows the Logic-Q model to be more risk-resilient by achieving much lower maximum drawdown under the market crash.

We then study the effectiveness of the Neural-TA module, we conduct a comprehensive weight distribution analysis by calculating the Rank-IC correlation between subpolicy returns and their ensemble weights. Formally, we compute $\mathrm{Rank-IC}(R_t, \phi_t)$, where Rank-IC represents the Spearman rank correlation coefficient (Spearman, 1904) between the return vector R_t and weight vector ϕ_t. Higher Rank-IC values indicate stronger correlation, suggesting that more profitable trading strategies receive proportionally higher ensemble weights, thereby reflecting superior decision-making quality. For rigorous comparison, we evaluate the original Logic-Q v0 model (employing Symbolic-TA only) against Logic-Q w/o LLM (denoted as *w/o LLM*, utilizing NeSy-TA), ensuring both configurations use identical Sharpe-Ens backbones. We conduct case studies across three representative market regimes for all evaluated markets: *volatile* (VOL) periods characterized by high price fluctuations, *descending* (DES) trends with sustained price declines, and *ascending* (ASC) trends featuring consistent upward movements. As demonstrated in Table 7, the *w/o LLM* model consistently achieves substantially higher Rank-IC values compared to the Logic-Q v0 across all three markets. These improvements stem from the more fine-grained market categorization of the Neural-TA module, which integrates neural components to capture subtle market patterns that the original pure symbolic approach cannot detect, enabling more precise dynamic weight allocation and ultimately leading to superior trading decisions.

Finally, to verify the robustness of the LLM module, we conducted comprehensive ablations as shown in Table 8. Replacing GPT-4o mini with stronger models such as GPT-4.1 and DeepSeek-V3 consistently delivers competitive or superior performance, with GPT-4.1 achieving the highest Sharpe ratio of 1.490 and cumulative return of 346.5%. We also tested three distinct prompt designs: professional trader style as our original prompt, risk-assessment style, and expert financial analyst style as shown in Fig. 7. The key differences lie in their role definitions and reasoning approaches. Our original prompt positions the LLM as a quantitative trader focused on direct sentiment analysis. Prompt 1 emphasizes risk evaluation from a financial risk analyst perspective. Prompt 2 adopts an expert analyst role that explicitly requires identifying key themes such as bullish/bearish signals and market catalysts before synthesizing recommendations. Despite these variations, all three prompts produce strong and comparable results across the evaluated measurements, with Prompt 3 achieving a Sharpe ratio of 1.365 and a cumulative return of 324.1%, confirming that our framework is highly robust to both model choice and prompt variation.

5. Related work

We briefly review two lines of research that are most relevant to our work: neuro-symbolic AI and AI for finance.

Neuro-symbolic AI Neuro-symbolic AI aims to combine symbolic reasoning with neural networks, leveraging the strengths of both paradigms. Symbolic systems offer structure, generalization, and interpretability, while neural networks excel at perception and learning from unstructured data. Their integration addresses limitations of each approach and has been successfully applied to tasks like reasoning, planning, and visual question answering. Early approaches pursued tight coupling between logic and neural learning. Logic Tensor Networks (LTNs) (Serafini & Garcez, 2016) encode first-order logic constraints in vector spaces, enabling differentiable constraint-based learning. Neural Theorem Provers (NTPs) (Rocktäschel & Riedel, 2017) and ∂ILP (Evans & Grefenstette, 2018) introduced differentiable reasoning over symbolic knowledge bases, facilitating interpretable rule learning through gradi-

Table 7
Weight distribution analysis of different methods.

Methods	USA			HK			CRYPTO		
	VOL	ASC	DES	VOL	ASC	DES	VOL	ASC	DES
Logic-Q v0	0.087	0.142	0.095	0.091	0.128	0.103	0.156	0.189	0.134
w/o LLM	**0.118**	**0.167**	**0.126**	**0.097**	**0.151**	**0.173**	**0.184**	**0.208**	**0.149**
Improvement	**35.6%**	**17.6%**	**32.6%**	**6.6%**	**18.0%**	**68.0%**	**17.9%**	**10.1%**	**11.2%**

(a) Prompt 1: Risk Assessment Style

```
System message: "You are a financial risk analyst evaluating market sentiment
    from news sources."

User message: "Given the recent news coverage for {ticker} over the past {
    time_window}:
[News 1: {Title} - {Description} ({Timestamp});
 News 2: {Title} - {Description} ({Timestamp}); ...],
assess the overall market sentiment and provide a single recommendation from the
    following options: Strong Buy (SB), Buy (B), Hold (H), Sell (S), or Strong
    Sell (SS). Respond only with: [Action]."
```

(b) Prompt 2: Expert Financial Analyst Style

```
System message: "You are an expert financial analyst with deep knowledge of
    market dynamics."

User message: "Review the following news about {ticker} published within {
    time_window}:
[News 1: {Title} - {Description} ({Timestamp});
 News 2: {Title} - {Description} ({Timestamp}); ...].
First, identify key themes (bullish/bearish signals, market catalysts, risks).
    Then, synthesize your analysis into ONE actionable recommendation: Strong Buy
     (SB), Buy (B), Hold (H), Sell (S), or Strong Sell (SS). Output only: [Action
    ]."
```

Fig. 7. Alternative prompt designs for LLM-based trading decisions.

ent optimization. Later models adopted modular architectures to improve scalability. The Neuro-Symbolic Concept Learner (NS-CL) (Mao et al., 2019) separates perception and reasoning by combining neural visual grounding with symbolic program execution, achieving interpretable reasoning capabilities. Relational RL models (Zambaldi et al., 2019) integrate explicit relational structures with deep RL, demonstrating improved generalization across related tasks. For knowledge-rich applications, models like COMET (Bosselut et al., 2019) infuse symbolic commonsense graphs into language models for enhanced reasoning. Lin et al. (2020) extend this approach with multi-hop relational reasoning over knowledge graphs. More recently, NeSy Transformers (Chen et al., 2023) embed symbolic priors directly into transformer architectures, reducing the computational overhead.

AI for finance The dynamic and increasingly volatile nature of financial markets has highlighted the limitations of traditional trading strategies, such as autoregressive moving average (ARMA) (Said & Dickey, 1984) and pair trading (Elliott et al., 2005). These approaches often struggle to capture complex non-linear relationships and adapt to rapidly changing market conditions. To address these challenges, the quantitative trading community has increasingly adopted deep learning and reinforcement learning techniques (Sutton & Barto, 1998; Wang et al., 2019), which have demonstrated superior performance in complex financial tasks including stock trading (Lim et al., 2019; Wu et al., 2020), order execution (Yu et al., 2020), and market making (Beysolow & Beysolow, 2019; Zhao & Linetsky, 2021). Among reinforcement learning approaches, value-based methods have gained particular attention for their effectiveness in handling discrete action spaces and market representations (Chen & Gao, 2019a; Deng et al., 2016; Zhang et al., 2020). The ability of these methods to learn optimal policies through trial and error makes them especially suitable for financial environments characterized by uncertainty and stochasticity. Various works have successfully applied Deep Q-Networks (DQN) and its variants (e.g., DRQN) to

Table 8
Ablation study on LLM model choice and prompt design for the US market under cash trading.

Model Ablation					
Methods	AR	CR	AV	MD	SR
Logic-Q (**Ours**)	21.2%	311.8%	16.1%	−21.8%	1.280
Logic-Q (GPT-4.1)	**22.8%**	**346.5%**	**15.3%**	**−20.1%**	**1.490**
Logic-Q (DeepSeek-V3)	21.9%	324.7%	15.8%	−21.2%	1.386
Prompt Ablation					
Methods	AR	CR	AV	MD	SR
Logic-Q (**Ours**)	21.2%	311.8%	16.1%	−21.8%	1.280
Logic-Q (Prompt 1)	20.6%	295.3%	16.4%	−22.5%	1.256
Logic-Q (Prompt 2)	**21.7%**	**324.1%**	**15.9%**	**−21.3%**	**1.365**

develop profitable trading strategies (Chen & Gao, 2019b), demonstrating their capacity to balance exploration and exploitation in dynamic market conditions. Building on these foundations, Li et al.propose LSRE-CAAN, a deep reinforcement learning framework for high-frequency portfolio management that addresses turnover constraints and quadratic dependency issues (Li et al., 2023). This approach minimizes transaction costs while maintaining competitive performance in volatile markets. In a similar vein, Duan et al.introduce an end-to-end DRL approach using probabilistic dynamic programming to optimize trading decisions across market graphs (Duan et al., 2022), which enables more comprehensive modeling of asset interdependencies. To improve optimization stability and reduce variance (Pricope, 2021), a persistent challenge in reinforcement learning applications to finance, Jiang et al.employ the twin delayed deep deterministic policy gradient (TD3) algorithm for risk-aware portfolio optimization (Jiang & Wang, 2022). Their approach incorporates risk metrics directly into the optimization objective, resulting in more robust trading strategies that perform well across different market regimes. Complementing these policy-based methods, Soleymani et al.develop a portfolio management framework based on a restricted stacked auto-encoder combined with a convolutional neural network to extract robust market features (Soleymani & Paquet, 2020).

6. Conclusion

In this paper, we propose a universal logic-guided DRL framework for quantitative trading, called Logic-Q, which embeds abstract human expert knowledge regarding market trends via the program synthesis by sketching paradigm. Logic-Q first introduces a neuro-symbolic trend analysis mechanism (NeSy-TA), which can dynamically determine the market trend and the corresponding tuning parameters for either the single-model reinforcement learning setting or the ensemble reinforcement learning setting, with the latter integrating backbone models from multiple modalities. Extensive experiments on two prevalent quantitative trading tasks demonstrate that Logic-Q significantly outperforms state-of-the-art baselines, including recent multimodal LLM-based trading agents. Regarding limitations, the neuro-symbolic trend analysis introduces additional computational overhead, potentially limiting applicability to ultra-high-frequency trading scenarios. More efficient implementations represent a key direction for future work. Additionally, we plan to extend Logic-Q to broader financial applications such as price prediction and market making.

CRediT authorship contribution statement

Junzhe Jiang: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization; **Zhiming Li:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization; **Aixin Cui:** Validation, Software, Data curation; **Xiao Zhou:** Writing – review & editing, Validation, Software, Resources, Investigation, Data curation; **Bo Li:** Writing – review & editing, Validation, Software, Resources, Investigation, Data curation; **Dongning Sun:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Investigation, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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